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Data Fusion and Information Extraction for Intelligent Integrated Pest Management

On the deduction and quantification of irruptive dynamics in mountain pine beetle population and proxy data

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Abstract

In attempting to develop a coherent description of the irruptive dynamics of mountain pine beetle (MPB) in North America, we examined a range of cases where authors described the dependency of intrinsic population growth rates on population attack levels. In some cases, detailed population data were used, but in most cases, investigators relied on operational aerial detection survey data. We found that study conclusions varied significantly depending on the type of data analysed and the spatial and temporal dimensions of the monitoring program. The ability to detect an intrinsic Allee effect (i.e., positive density-dependent growth for low and rising population densities) depended on the type and the intensity of the sampling effort and the extent of sampling in space and time. Consequently, not all studies were able to quantify the irruption threshold (i.e., the population density at which endemic populations may transition towards the epidemic state). Notably, in every study where an Allee effect was demonstrated, investigators also identified at least one extrinsic environmental factor (e.g., winter weather, summer drought, microclimatic effect) that was regulating its strength. Our results suggest that population surveys conducted from the ground are a necessary complement to aerial survey data if the goal is to make inferences about the irruptive potential for MPB populations and the role of environmental factors in shaping that irruptive potential.

KEYWORDS

Allee effect, irruptive dynamics, mountain pine beetle, population modeling

1 | INTRODUCTION

The rediscovery of the Allee effect (Allee, 1931) by ecologists and conservation biologists in the 1990s led to an outbreak of studies searching for evidence of the effect in numerous animal and plant systems. This intrinsic population effect is assumed, or theorized, to govern the dynamics of many animal systems and insect systems in particular (Watt, 1960); however, evidence is equivocal and either

considered to be conclusive (Kramer et al., 2009) or sparse (Gregory et al., 2010) depending on how the empirical data are analysed. Much of this ambiguity stems from the fact that the intrinsic Allee effect operates at low densities, where populations are often difficult to locate or study effectively (Kramer et al., 2009). As the forest health sector continually innovates in the area of digital acquisition of pest data, a critical question must be answered: What is the standard to be met if the goal is the early identification of eruptive pest risks?

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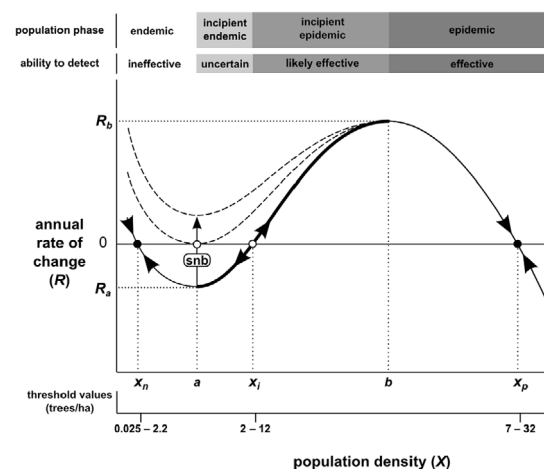
The mountain pine beetle (MPB), *Dendroctonus ponderosae* Hopkins (Coleoptera: Curculionidae), is one of the most influential insects in North American forest ecosystems (Kurz et al., 2008; Raffa et al., 2008). The insect derives its influence from the significant tree mortality it can cause during outbreaks. This mortality affects a range of ecological, economic and social processes and has spurred research efforts towards understanding the factors that influence the population dynamics of this species. The result has been good support for the hypothesis that the population dynamics of MPB are influenced by component Allee effects (Stephens et al., 1999) that are the result of the resin-based defence system employed by pine trees to protect themselves from beetle attack (Nelson & Lewis, 2008; Raffa et al., 2005). The defence system mediates the Allee effect in MPB populations because most tree species attacked by the beetle have pre-formed (i.e., constitutive) cortical resin ducts that when severed release resin capable of expelling, or 'pitching-out', attacking beetles and inducible defences derived from toxic resins produced at the site of attack that rapidly increase in concentration to levels that exceed the physiological tolerance of beetles (Franceschi et al., 2005; Raffa et al., 2005). These defences are most influential when MPB populations are small because they prevent most attacks from succeeding. Thus, populations are restricted to exploiting dead and dying trees (Howe et al., 2022; Safranyik & Carroll, 2006) that lack a functioning defence system (Boone et al., 2011). However, when MPB populations are large, individual beetles can overwhelm a tree's defence system if they combine their efforts with large numbers of other individuals. This combination of efforts is referred to as a 'mass attack' and occurs when many beetles attack the same tree at the same time. This mass attack process is itself mediated by the pheromone biology of female and male beetles, their physical and chemical interaction with the tree and the tree's internal defence system (Raffa & Berryman, 1983). Once populations surpass a threshold density required for cooperative attacks, an abrupt population irruption may follow if there are sufficient hosts over a landscape. The overall result is a complex process, but one in which the effect can be observed in the individual reproductive rate. Initially, the individual reproductive rate rises with population size, peaks at a point of intermediate attack density and then declines as population size increases even further. The decline occurs either because competition among larvae for food (phloem) limits the growth of populations (e.g., Lewis et al., 2010; Raffa & Berryman, 1983; Reid, 1963), or because of emigration from a stand undergoing significant host depletion, such that host-limitation promotes dispersal competition for better habitat (Aukema et al., 2008; de la Giroday et al., 2011; Powell & Bentz, 2009).

The ecological knowledge regarding individual MPB reproduction on individual trees has been used to hypothesize a theoretical growth curve for MPB populations (Berryman, 1979). This model describes the behaviour of an organism with a strong demographic Allee effect (Taylor & Hastings, 2005). A generalised theory of eruptive outbreak via the Allee effect is accordingly outlined in Box 1.

The Allee effect is diagrammed in Box 1 as the thick-shaded, positively sloped portion of the recruitment curve. This region describes the incipient behaviour of a MPB population as it transitions from an

BOX 1 The dynamics of population irruption from endemic to epidemic via saddle-node bifurcation, and implications for the ability to detect the underlying Allee effect

Arrows on the recruitment curve indicate the direction of population change amongst equilibrium states. Below $R=0$, populations decline, above $R=0$, populations rise. At $R=0$, populations do not change. Circles indicate thresholds where populations transition between rising and declining states, solid circles indicate stable thresholds and the open circle is an unstable threshold (X_n =endemic threshold, X_i =incipient threshold, X_p =epidemic threshold). The Allee effect is indicated by the thickened portion of the curve. It spans the interval (R_a , R_b), leading to a positive correlation between the annual per capita rate of change and population density over the interval (X_a , X_b). The vector 'snb' indicates a saddle-node bifurcation that will lead to an outbreak if the endemic portion of the recruitment curve is shifted upwards, resulting in the dynamics shown in the broken curves. Shaded bars above the main plot indicate the divisions of the population dynamics into phases (upper bar) and the effectiveness of proxies to detect these dynamics using tree mortality data (lower bar). The secondary x-axis below the main plot gives empirical values for the thresholds obtained from the MPB literature.



endemic phase, when it is restricted to dead and dying trees, to an epidemic phase when it can exploit well-defended trees; a density-dependent process that is associated with a shift in host preferences by attacking beetles (Boone et al., 2011; Burke & Carroll, 2017). The Allee region of the growth curve can itself be split into two segments at the incipient threshold (X_i) which we label as the incipient endemic phase and the incipient epidemic phase. Note the incipient population "phases" are, by our definition, not discrete states, but rather regions around the irruptive threshold whereby populations may be driven to either the endemic or epidemic states as a result

of a perturbation. This perturbation can take the form of either a change in density, leading to movement along the trajectory of the recruitment curve, or a change in recruitment for a given density, which changes either the position or shape of the recruitment curve.

This characterisation of the Allee effect in MPB populations is rooted in the idea that aggregated populations exhibit higher survival rates than non-aggregated populations. Specifically, in bark beetles the upward-sloping “cooperation curve” is hypothesized to occur when larger aggregations of beetles, by cooperating in the defeat of host defences, fare better than they would otherwise (Berryman, 1979; Howe et al., 2022). Our definition of the incipient phase is an extension of Safranyik and Carroll's (2006) who defined the incipient-epidemic phase of an MPB population as “one where MPB populations have grown to a minimum size sufficient to successfully mass attack a single large-diameter tree within a stand”. Such populations are expected to grow slowly in the initial few years, but as long as they do not fall below X_i they will eventually increase in size, as will the number of trees that are attacked (Safranyik & Carroll, 2006).

The evidence for a demographic Allee effect and irruptive Allee dynamic, is well-established for MPB at the level of individual trees in controlled settings (e.g., Raffa & Berryman, 1983). However, evidence that Allee dynamics play a role in operational settings at the stand and landscape scales is rare. At the stand scale, Allee-driven dynamics were purportedly detected in data from Montana (Berryman, 1979, 1999) but not detected in a very similar data set from British Columbia (Trzcinski & Reid, 2009). A combined analysis that used data from those two studies and others (e.g., McGregor et al., 1987) also did not detect evidence of Allee dynamics (MacQuarrie & Cooke, 2011). At the landscape scale, the evidence for Allee effects is equivocal. For instance, populations in Idaho (Heavilin & Powell, 2008) were better fit by a Holling type III (sigmoidal) model than a saturating type II model (Holling, 1959), which suggests Allee effects might be operating at the landscape scale. However, other work from the same region showed that Allee effects are not required to explain tree mortality in an outbreak that began in the 1990s if climatic variability is included as a driver in the analysis (Powell & Bentz, 2009).

If we accept the premise that demographic Allee effects are ubiquitously important to the dynamics of MPB populations and that effects observed at one scale ought to be present at all scales, then we are led to ask: where is the evidence for Allee effects in MPB in stand- and landscape-scale data? We hypothesized that the lack of evidence may be due to the low intensity of sampling efforts in the majority of studies.

Many studies of MPB outbreak dynamics are conducted using tree mortality data derived from operational surveys (which we refer to as ‘proxy data’, as tree death is effectively a proxy for population colonisation success). The ability of these aerial surveys to detect freshly attacked trees has never been formally examined; however, it is known that aerial surveys are fallible on at least two grounds: (1) trees fade slowly from green to red as they die and the earlier the survey, the lower the probability of detecting fresh attacks from the air (Wulder et al., 2006); (2) operational surveys in a region are often initiated a few years or more after the first red attack clusters appear (Safranyik & Carroll, 2006). Indeed, the fallibility of aerial

surveys is the major reason why ground surveys are often used as a follow-up measure for obtaining more detailed observations of MPB attack levels and attack success. Consequently, we were interested in survey effort between the endemic and irruptive phase (i.e., survey timing) as a factor influencing the suitability of the tree mortality data as a proxy for population colonisation success.

We undertook a detailed and comprehensive literature review, with associated data re-analysis, to determine if the strength of the evidence supporting the existence of Allee effects in the MPB system at scales broader than individual trees was correctly stated and whether it related to: (1) the type of data gathered; (2) the timing and duration of sampling and (3) the spatial extent of sampling. We predicted that the more intensive the field study, the higher the probability of detecting an intrinsic Allee effect, which is defined as the co-operative effect of rising beetle population density on rates of population growth in the vicinity of the unstable irruption threshold. We further predicted that those studies that were well-positioned to observe an intrinsic Allee effect also had a greater chance of detecting a strong effect of extrinsic environmental factors on population irruptive potential.

The ultimate goal of this study was to explore the possible connection between operational data streams and intelligent predictive pest management, by attempting to derive a simple, coherent description of the basic, intrinsic, irruptive dynamics of MPB, such as that represented in the conceptual model of Box 1. Our specific questions were as follows:

1. Does the MPB exhibit irruptive behaviour, characterised by a positive density-dependent Allee effect?
2. If an Allee effect is present, is it strong enough to create an unstable equilibrium point that would behave as an irruptive threshold?
3. What is the estimated irruption threshold and how does it vary?
4. Does the irruption threshold occur at a population density that is below the detection threshold inherent in traditional operational detection surveys?
5. How does the maximum reproductive rate vary across studies and environmental conditions?
6. What is the role of environmental factors in shaping the irruption threshold?

Answers to these questions have direct operational significance because the existence of an Allee effect implies that there is some low rate of attack below which it may be safe to leave localised infestations to self-extinguish. Conventional tactics for the management of irruptive bark beetles such as MPB comprise the destruction of infested trees through harvesting and processing or felling and burning; however, this approach is costly, labour-intensive and logistically difficult when infestations occur in remote areas (Carroll et al., 2006). The ability to “walk away” from low-density attacks that pose no threat allows control efforts to target populations that are most likely to contribute to irruptive growth and spread. Similarly, if epidemic infestations occur in areas of unsuitable host or climate, where populations are highly unlikely to irrupt and spread, then significant cost savings may accrue by excluding those areas from treatment, with no loss of benefit.

2 | METHODS

2.1 | Literature survey

We examined a wide range of field studies from the literature where investigators attempted to describe or model a relationship between MPB population growth rates and MPB infestation size. Studies were characterised in terms of (1) the method of survey used to obtain the data (ground survey of populations versus aerial survey of tree mortality), (2) the timing (early versus late) and duration (prolonged versus single-year) of the monitoring program, (3) whether or not evidence for an Allee effect was reported and (4) whether there was convincing evidence for a shift in irruption dynamics related to an extrinsic environmental variable.

We predicted that the intrinsic dynamics of irruption from the endemic state would not be visible in survey data unless the surveys were:

1. Initiated early, well before the “sudden” appearance of attack clusters (such as a central mass attack with ancillary spill-over attacks);
2. Intensive, with the ability to detect freshly colonised trees occurring at a rate lower than one tree per hectare;
3. Conducted across plots that are sufficiently large that when the first red tree is detected, it will be accompanied by other attacked trees not successfully mass-attacked;
4. Conducted on the ground and after aerial survey, to determine, at minimum, the number of trees attacked unsuccessfully (which are trees that do not fade to red and are not aerially detectable).

We predicted that these conditions are rarely met in operational circumstances, but are occasionally met when professional population ecologists are leading the study. Finally, we predicted that the ability to identify an intrinsic Allee effect was a necessary precondition for reporting a quantified effect of an extrinsic environmental factor on irruption dynamics.

2.1.1 | Field study of modern outbreak

The sudden emergence of a new MPB outbreak in northern British Columbia in the early 2000s provided a novel opportunity to add to the body of historical observations involving ground surveys. The outbreak was monitored in two locations, near Tumbler Ridge, British Columbia, in the summer of 2006 and in Grande Prairie, Alberta, in the summer of 2007. We expected that these populations, as a result of massive immigration, were already beyond the outbreak threshold at the time of sampling, such that no Allee effect would be visible in plots of reproduction versus attack density.

At each of the two locations, ten candidate stands of pure lodgepole pine were visited based on reconnaissance using aerial survey data. Six stands were selected at Tumbler Ridge for study and 8 stands at Grande Prairie (Table 1). Within each stand, a one-hectare block was identified for intensive population study by ground surveys. Within this one-hectare survey block, a standard prism sweep, with a basal area factor of 5, was conducted and MPB populations were sampled using under-bark samples. For each tree falling into the prism sweep, injuries putatively associated with reduced vigour and defensive capacity was recorded (i.e., prior stressors; Howe et al., 2022). The total number of trees with current MPB attacks in each block ranged from 13 to 197. Two samples were taken from each attacked tree, one from the north side and one from the south side, at a height of approximately 1.3 m above ground. Samples comprised squares of bark (15 × 15 cm) cut from bole with chisels. Within each square, all eggs, larvae, pupae and adults were counted. MPB attack density was measured as the number of female attacks per m² of bark surface area. Reproduction was calculated as the ratio of juveniles to female adult attacks within each sample. These sampling protocols are standard experimental procedures and have been documented in detail elsewhere (Carroll et al., 2006; Safranyik, 1968). Finally, attack densities and reproductive rates were binned by attack

Region	Site no.	Site label	Longitude (°W)	Latitude (°N)
Tumbler Ridge BC 2006	1	Bullmoose 1 Gas well site	121.4038667	55.2117167
	2	Bullmoose 2 Main Road	121.2801333	55.2199167
	3	Monkman 3	121.1499667	54.7750667
	4	Monkman 4	121.2190167	54.8404333
	5	Moose Lake 5 East	121.1291167	55.2742833
	6	Moose Lake 6 West	121.2911500	55.2334167
Grande Prairie AB 2007	1	Dunes 1	118.7853333	55.0819500
	2	Dunes 2	118.7169500	55.0822167
	3	Grande Prairie	118.7002667	55.0336167
	4	Wapiti	119.0650000	55.0302667
	5	Bald Mountain	118.9555500	54.8302500
	6	Economy Creek	118.3641667	54.8522000
	7	Imperial	118.5741667	54.6850000
	8	Greas	118.5058333	54.6944500

TABLE 1 Sample locations for the 2006–07 MPB population study.

density and averaged within each equally sized bin. Eleven data bins were used for the Tumbler Ridge 2006 dataset. Seven data bins were used with the Grande Prairie 2007 dataset.

2.1.2 | Case studies comparative analysis

The studies identified in the literature review were highly idiosyncratic in terms of choice of the study area, types of observations gathered, survey procedures and parameters estimated, so rather than conduct an *a priori* statistical meta-analysis, we did a qualitative, post hoc, comparative examination of how the conclusions in these studies might have been affected by the quality of the source data used in the analysis.

For each of the studies reviewed, we obtained, where possible, the raw data to develop our models of the relationship between population growth rates and attack densities, to verify any claims of the authors *vis-a-vis* the existence of positive feedback between population growth rates and attack densities that would influence the likelihood or severity of irruptive behaviour. We used these data to estimate the endemic threshold (if it existed), the irruption threshold (if it existed) and the maximum population growth rate (if it existed) – all as defined in [Box 1](#). We compiled notes on factors that were demonstrated to affect these parameters. All data transformations and analyses were conducted using the R statistical software package ver. 4.2.3 (R Core Team, 2023). All data and code have been publicly archived at <https://doi.org/10.13140/RG.2.2.25960.65283>.

3 | RESULTS

3.1 | Field study of modern outbreak

Sampling of the MPB outbreak populations at Tumbler Ridge, British Columbia in 2006 and Grande Prairie, Alberta in 2007 revealed a range in attack densities and reproductive rates. The lowest attack densities observed were ~16 galleries/m² and the highest were ~250 galleries/m². Population rates of change declined from 30-fold to 4-fold with increasing attack density, with no sign of rising recruitment at the lower attack densities. These growth rates are consistent with an interpretation that surveys were initiated when populations were already beyond the irruption threshold. Results from this population study were subsequently folded into the meta-analysis of historical data, where they are presented in context (next section).

3.1.1 | Population and detectability thresholds in proxy studies

Empirical estimates for the endemic and irruption thresholds were only given by a few of the proxy studies of tree mortality. These

are summarised in [Table 2](#). Amman (1984) suggested an endemic population comprised ≤ 1 mature tree killed per 100 acres (40.5 hectares), or 0.025 trees/ha (all estimates have been converted and expressed as trees per hectare). These parameters were referred to as “guesstimates” that were based on the authors' experience. Berryman (1979) originally estimated an endemic threshold of 0.002 trees/ha. However, Berryman (1999), using a different model on the same data, revised that estimate upward three orders of magnitude to 2.2 trees/ha. Safranyik and Carroll (2006) employed a behavioural definition, reporting an endemic population as attacking 3–6 trees/ha/y at a rate of just 2–3 attacks per tree, with no successfully mass-attacked trees occurring at any scale. These authors distinguished between successfully mass-attacked mature trees, defined as the “epidemic niche” ($X_n = 0$) and sparse attacks on suppressed trees ($X_n = 3–6$), defined as the “endemic niche”. Based on evidence of spatial aggregation within stands, Howe et al. (2022) estimated the endemic threshold at 1–2 trees/ha.

For the incipient threshold, we found estimates of 12 mature, mass-attacked trees/ha (Amman, 1984) and 5 trees/ha (Berryman, 1999), with the latter replacing the original estimate (Berryman, 1979) of 0.032 trees/ha. Howe et al. (2022) estimated the threshold at 2–4 trees/ha. Safranyik and Carroll (2006) described an irruptive Allee threshold but did not define it quantitatively, noting only that this threshold depends greatly on tree diameter distributions within the forest.

Given the disagreement between Berryman (1979, 1999), we re-analysed those data using a hybrid third model – one that allowed for both nonlinear recruitment dynamics (as in the 1979 paper) and also differentiated between endemic and epidemic niche recruitment dynamics (as in the 1999 book chapter). We derived endemic, irruptive and epidemic thresholds that were more comparable, but not identical, to Berryman (1999) ([Table 1](#)). [Figure 2](#) illustrates the original data of Tunnock (1970) and Parker (1973) (panel a), along with Berryman's (1979) hand-drawn derived recruitment curve (panel b). [Fig. 3a](#) illustrates our best attempt to recreate Berryman's (1979) curve and his (1999) multi-niche model, showing clearly that the dynamic equilibrium points (endemic, incipient, epidemic) that we computed do not at all match those of Berryman (1979). We did, nevertheless, find support for the existence of a strong Allee effect in the original field data of Tunnock (1970) and Parker (1973).

These results found in the earlier literature are complemented by estimates from a more synthetic model of MPB irruptive dynamics in southcentral British Columbia during a recent outbreak (Cooke & Carroll, 2017). That study identified the endemic, incipient and epidemic thresholds as 1.0, 2.5 and 47 trees/ha/y. According to this model, these thresholds are predicted to shift as forest conditions and climatic conditions vary, with the maximum rate of reproduction varying between 1.8-fold and 4.5-fold. The parameters, listed in [Table 2](#), are taken from Cooke and Carroll's (2017) figure 7a. The range on $\max(R_p(X))$

TABLE 2 Threshold estimates drawn from the literature of proxy-based studies of MPB growth.

Reference	Estimated threshold parameter			
	X_n trees/ha/y	X_i trees/ha/y	X_p trees/ha/y	Max($R_p(X)$)
Berryman (1979)	0.002	0.032	11.2	12.6
Berryman (1999)	2.2	5	12.3	Linear/NA
Our figures 2/3a re-analysis of Berryman (1979)	1.5	8.5	36.7	1.86
Amman (1984)	0.025	12	-	-
Safranyik and Carroll (2006)	0, 3–6	-	-	-
Cooke and Carroll (2017)	1.0	2.5	47	1.8–4.5
Howe et al. (2022)	1–2	2–4	-	-

represents the range estimated for cool to warm climates in southcentral British Columbia.

The literature estimates for the endemic threshold (Box 1) bracket a region where it would be necessary to survey a large area to detect individual attacks (e.g., at least 40ha for the low guesstimate, assuming this is large enough to contain the 3–6 suppressed trees stipulated by Safranyik & Carroll, 2006). Less effort would be required if the threshold is closer to the upper value of 2.2 trees/ha. Regardless, it is in the endemic phase that important aspects of the beetle's biology come into play that make surveying difficult. Small populations of MPB tend not to attack large, healthy trees. Instead, they preferentially attack weak, impoverished hosts behaving much like secondary bark-beetles (Boone et al., 2011; Howe et al., 2022; Safranyik & Carroll, 2006; Smith et al., 2011). However, since tree mortality surveys are concerned with the loss of economically valuable timber, it is the large, healthy trees that are commonly the subject of mortality surveys (e.g., McGregor et al., 1987). We conclude that detecting the behaviour of endemic populations in proxy data is largely intractable when using tree mortality as a proxy for population size because these trees will be rare on the landscape and not of the type that would be included in a standard survey.

Based on the literature estimates for the irruptive threshold (2–12 trees/ha), we suggest it is reasonable to predict that Allee effects can be detected in plots as small as 1 ha. Assuming that a single tree killed by MPB could be found with a determined search effort within an arbitrary stand density of 1000 trees/ha, it should be possible to detect dead trees efficiently enough to parameterise the incipient epidemic phase of the growth curve (Box 1). Similarly, we should also be able to characterise the rightmost portion of the incipient endemic phase (i.e., where values are slightly <5 trees/ha) without much extra effort. However, when populations are smaller (near a ; Box 1) beetles are predicted to be moving from defensively impaired hosts to well-defended hosts (Boone et al., 2011; Safranyik & Carroll, 2006). These mass-attack attempts are predicted to fail and not kill trees, or if they do kill trees they are expected to be rare on the landscape. Thus, as we argue above, these attacks would have a high probability of going undetected in an aerial survey. The result would be a set of tree mortality data that is less reliable as a population proxy when the population is in the early portion of the incipient phase (MacQuarrie & Cooke, 2011).

3.1.2 | Metadata analysis

We identified nine empirical studies where investigators attempted to model the relationship between population growth rates and attack densities based on field data, the original field data reported here constituting a tenth (Table 3). The studies that we identified varied considerably in terms of the method of survey (ground survey of populations versus aerial survey of tree mortality), the timing (early vs. late) and duration (prolonged versus single-year) of the monitoring program and whether or not evidence for an Allee effect was detected. Three of the studies (Berryman, 1979; Cudmore et al., 2010; Waring & Pitman, 1985) were conducted in disjunct regions of western North America (Montana & Wyoming, Oregon and British Columbia; respectively). Four of the studies (Boone et al., 2011; Howe et al., 2022; MacQuarrie & Cooke, 2011; Trzcinski & Reid, 2009) intersected in the vicinity of southern British Columbia and could be loosely compared based on geography and contrasted based on survey methods. Two of the studies were conducted in Idaho on the same data set, with Heavilin and Powell (2008) analysing the data spatially, year-by-year and Powell and Bentz (2009) analysing the same data as a longitudinal time series of aggregated spatial data. Of these ten studies, only one (Cudmore et al., 2010) quantified recruitment across the full range in attack density.

Results in Table 2 supported our intuition that a wide range of survey methods would lead to variation in the strength of the evidence supporting the existence of an Allee effect. The studies could be split into two distinct categories, based on the strength of evidence supporting the existence of an Allee effect. Five of the ten studies reported an Allee effect; five did not. We plotted the data from each of these studies to illustrate the presence or absence of Allee effects reported in Table 3 (Figure 2). Evidence from the ninth study, from Cudmore et al. (2010) presented a split case from British Columbia, with three of the four study plots ('Prince George', 'Rocky Mountain', '100 Mile House') being post-irruptive and one ('Columbia') sampled early enough that it was still mid-irruption, the outbreak having peaked in the North before the South (Aukema et al., 2006). The Columbia study area was climatically much cooler than the other three – a likely explanation for its slower progression towards outbreak – so we split the

TABLE 3 Population and proxy-based studies of MPB attack rates and consequences of attack.

Study #	Authority	Location	Years	Extent (km ²)	No. samples	Area per sample (ha)	Survey type	Datum	Spatial aggreg? ^a	Early?	Long?	Allee?	Extrinsic factors
1	Berryman (1979) ^b	MO-WY	1956–1972	8	1	800	Aerial	Tree mortality	y	n	y	y	Aspect, microclimate
2	Waring and Pitman (1985)	OR	1979	0.0756	16	0.25	Ground	Population attack success	n	y	n	y	Host vigour
3	Heavilin and Powell (2008)	ID	1989–2004	1080	1.2 × 10 ⁶	0.09	Aerial	Tree mortality	n	n	y	y	Forest density, weather, microclimate
4	Powell and Bentz (2009)	ID	1989–2004	1080	1	1080	Aerial	Tree mortality	y	n	y	n	Weather, microclimate
5	Trzcinski and Reid (2009)	S. BC	1931–1943	15	7	0.4	Ground	Tree mortality	y	n	y	n	Weather; microclimate. aspect
6	Carroll (this study)	N. AB-BC	2006–2007	2000	14	1	Ground	Population brood production	n	n	n	n	NA
7	MacQuarrie and Cooke (2011) ^b	Central Rockies	1936–2004	NA	NA	NA	Aerial & ground	Tree mortality	NA	n	y	n	Forest density
8	Boone et al. (2011)	S. BC	2000–2005	800	6	15	Ground	Population attack success	n	y	y	y	Host vigour
9 ^a	Cudmore et al. (2010)	N. BC	2005–2006	3000	11	10	Ground	Population brood production	n	n	y	n	Host vigour, climate
9 ^b	Cudmore et al. (2010)	S. BC	2006	1000	3	10	Ground	Population brood production	n	y	n	y	Host quality, climate
10	Howe et al. (2022)	S. BC	1999–2005	800	6	15	Ground	Population attack success	n	y	y	y	Host vigour

^aSome studies report only aggregated dead tree counts at the level of the entire study area. At the other extreme, the more detailed studies record thousands of observations in a nested hierarchy of galleries within stems with plots within study area and do not aggregate the data in statistical analysis. This variable attempts to split the studies into two categories based on this degree of data richness. This study was a re-analysis of an amalgamation of eight historical tree mortality datasets spanning the central Rocky Mountain region of the US and Canada, including studies #1 and #5. The area sampled is larger than any other study listed here, but the sampling scheme is difficult to characterise because of variations in individual sampling methods used in each sub-study. Data for this analysis were aggregated at the level of the sub-study, but not higher.

Cudmore et al. (2010) study into two parts, based on climatic suitability. The tenth study (Howe et al., 2022) re-examined the data of Boone et al. (2011), but in a spatial analytical context and added year of data. Howe et al., 2022 reached the same conclusion about the presence of the Allee effect and mediation by environmental factors, so to avoid double-counting, we chose not to include their data in fig. 3.

Of the five studies that did not report or display an Allee effect (fig. 3f–j), three were based on operational time-series data for tree mortality, aggregated across a large study area. A fourth, (Results section 1.0) described here in Methods section 2, was conducted late in the outbreak sequence (2006–07) in a northern region East of the Rocky Mountains, when immigration was heavy and low-density attacks were exceedingly rare. A fifth, the split result from Cudmore et al. (2010), involved populations that were well beyond the irruptive phase, attacking trees with compromised defences.

Of the five studies that did report an Allee effect (fig. 3a–e), two were based on stem-level experimental-quality population monitoring data that were not aggregated across a large study area. A third was the only study to report an Allee effect based on operational aerial survey time-series data. However, we question that result, as we were not able to duplicate that model result (see section 4.1). A fourth was the only study to indisputably report an Allee effect based on operational aerial survey data and this was achieved using a spatial model where the statistical model failed to converge in 4 out of the 12 years during which data were available for model parameterisation. The fifth was one of the split results from Cudmore et al. (2010), where survey intensity was high and because of cool climatic conditions, population trajectories lagged behind those in the core outbreak area of most of British Columbia.

Of the three population monitoring datasets that reported an Allee effect, Waring and Pitman (1985) observed an endemic–epidemic transition spatially, within a single year, while Boone et al. (2011) observed an endemic–epidemic transition temporally, across years. These two studies were conducted early enough in the outbreak sequence and over a large enough area, that there was high spatial variation in population attack levels. Here, the effect of per-stem attack density on per-stem population growth rate could be analysed because data were not aggregated across the entire study area. Howe et al. (2022), for example, showed how a spatially explicit treatment of the individual stem data from Boone et al. (2011) could produce a more robust result. This speaks to the value of highly resolved spatial data collected over wide enough areas that a broad range in attack density is spanned. If low-density attacks can be found and tree defences are not compromised, then an Allee effect will be detected.

In every case where an intrinsic Allee effect was identified, investigators also quantified the effect of an extrinsic factor that was shaping the endemic and incipient thresholds governing irruption probability. Waring and Pitman (1985), in Oregon, clearly demonstrated the role of tree vigour in mediating the probability that either a tree or the attacking beetle population survives a

mass attack. Heavilin and Powell (2008), in Idaho, showed that the strength of the type III Holling response curve used to describe the ability of attacking beetles to successfully colonize a tree was determined by climatic conditions in cooler years (1990–1995) versus warmer years (1997–2001). Cooke and Carroll (2017) showed that a year-by-year breakdown of the response curve of Boone et al. (2011) in British Columbia – the probability of successful tree colonisation as a function of the number of trees already attacked – revealed a significant difference in the shape of the response curve depending if the year was dry or wet. Two of the studies that did not illustrate an Allee effect of positive density-dependence – Trzcinski and Reid (2009) in British Columbia and Powell and Bentz (2009) in Idaho – nevertheless reported that extrinsic climatic and microclimatic factors were significantly involved in shaping the negative density-dependent dynamics of population growth post irruption.

3.1.3 | Case studies comparative analysis

US National Parks

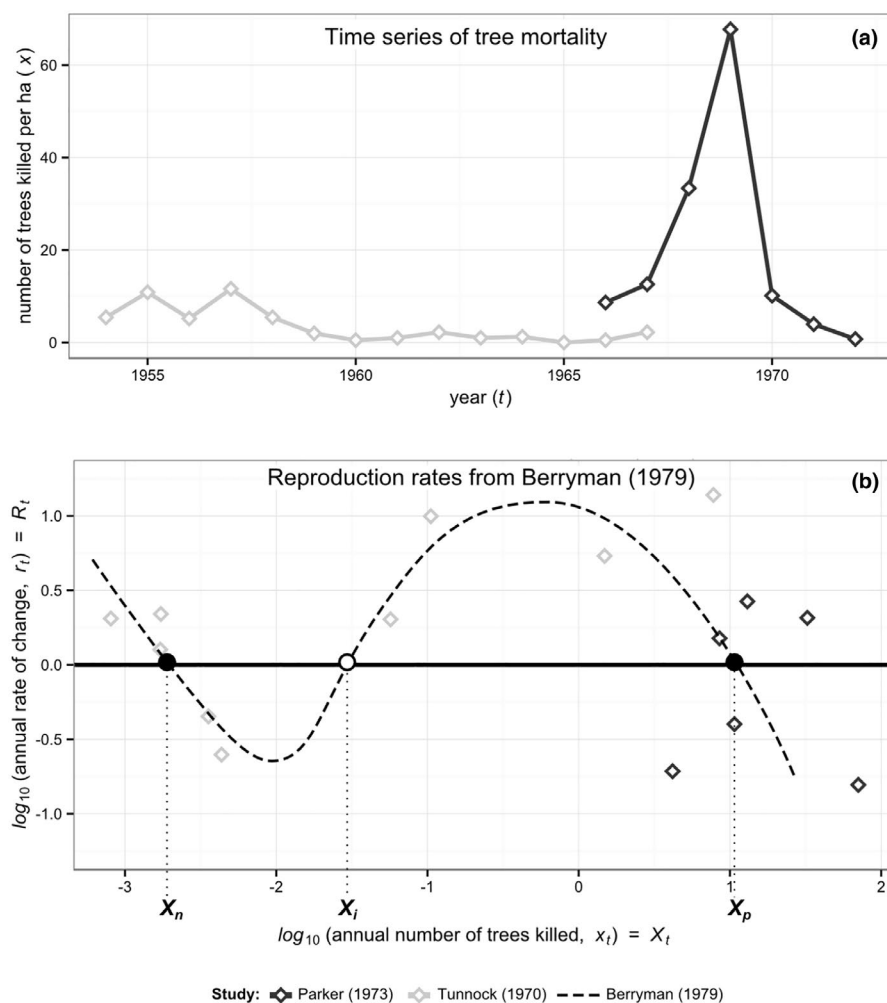
Tunnock (1970) and Parker (1973) reported tree mortality data for Glacier National Park in Montana and Yellowstone National Park in Wyoming (Figure 1a) that was later used by Berryman (1979, 1999) to derive a population curve for MPB (reproduced in Figure 1a,b). With these data, Berryman (1979, 1999) constructed a recruitment curve with a strong demographic Allee effect (Figure 1b). Our version of a recruitment curve constructed using the same data is illustrated in Figure 2a.

While our derived curves (Figure 2a) do not exactly match those of Berryman (Figure 1b) they do indicate the same multiple-equilibrium dynamic structure as that hypothesized in Box 1. Note that we attempted to reproduce Berryman's curve using the same data but were unable to do so, even after following the recommendation to adjust the chronic phase (epidemic phase) and endemic-phase time series to a common mean (Berryman, 1999). Regardless, that we can find three equilibria in these data suggests that proxy tree mortality data can be used to construct the entire reproduction curve for an MPB population. Moreover, the values of these equilibria ($X_n = 1.5$ trees/ha; $X_i = 8.5$ trees/ha, $X_p = 36.7$ trees/ha) fit well with the ranges predicted by the literature (Box 1). Further inspecting these data supports the model's prediction that MPB populations are characterised by a strong demographic Allee effect (as in Taylor & Hastings, 2005) that would govern the occurrence of population irruptions.

Kootenay national park

Hopping and Mathers (1943) reported a long series of tree mortality observations from nine stands in Kootenay National Park in southern British Columbia. Trzcinski and Reid (2009) used data from seven of these stands to parameterize a reproduction curve for MPB and examine the roles of environmental factors on population growth. They did not observe an Allee effect in the Kootenay National Park

FIGURE 1 Tree mortality and derived reproduction functions for mountain pine beetle populations in two US National Parks (Parker, 1973; Tunnock, 1970). (a) time series of tree mortality from each study; (b) annual rate of change and reproduction function re-plotted from Berryman (1979) which is based on the data in (a) after “converting to a common mean” – a data manipulation that is unclear and could not be reproduced to generate curve (b) from the data in (a). Our best attempt to recreate Berryman's (1979, 1999) model of these data is illustrated in fig. 3a.



data, which they attributed to the fact the MPB outbreak they studied was preceded by a drought. Drought decreases the resistance of trees to MPB (Thomson & Shrimpton, 1984) which can permit MPB to 'jump' from the endemic phase to the epidemic phase. This occurs because of the relaxation of the Allee effect such that population function shifts upwards giving a positive growth rate for all population densities (Figure 1). Under these conditions, it might not be possible to observe a strong Allee effect because the conditions that enforce the effect are temporarily relaxed. Moreover, the shift to the epidemic phase is rapid, (e.g., 1–2 generations) and can be easily missed.

As an aside, we point out that Richmond (1936), reporting on the Kootenay outbreak, attributed the infestation to emigration from a larger outbreak south of the park, not drought. This immigration event resulted in a population irruption in most of the surrounding valley and seven of the stands simultaneously (Richmond, 1936). The remaining two stands were infested a few years later (Hopping & Mathers, 1943). Regardless, of whether it be drought or mass immigration, the effect on the population growth curve would be the same – a temporary relaxation of the Allee effect allowing populations to escape the endemic phase without passing through the portion of the incipient phase with negative per-capita growth rates.

This case study illustrates conditions under which an Allee effect may exist but can be overcome by a shift in the location of the reproduction function. This type of shift is described as a 'saddle-node bifurcation' (Berryman, 1982) that allows endemic populations to proceed to outbreak when populations exceed X_i , either by immigration or by a shifting of recruitment curves resulting in the collision and subsequent disappearance of X_n and X_i ('snb' in Box 1). As evidence, in four of the seven Kootenay Park plots populations dropped as low as 2.5 trees per hectare (Hopping & Mathers, 1943). This suggests that survey efforts through this phase were sufficient to detect very low attack densities of one tree per hectare or less. Indeed, in the study, every tree in each plot was tagged and revisited annually (Hopping & Mathers, 1943; Richmond, 1936). Based on this evidence, we propose that if MPB was endemic in the nine Kootenay Park stands prior to 1935 and if the rise to the epidemic phase after 1936 was the result of a process endogenous to those stands, then an Allee effect should have been detected because the stands were being surveyed at sufficient resolution to produce observations that would reveal the effect. Since an Allee effect is not present in these data (MacQuarrie & Cooke, 2011; Trzcinski & Reid, 2009) it would suggest that another process exogenous to the stand was responsible. Immigration (Richmond, 1936) would seem to be the most parsimonious explanation.

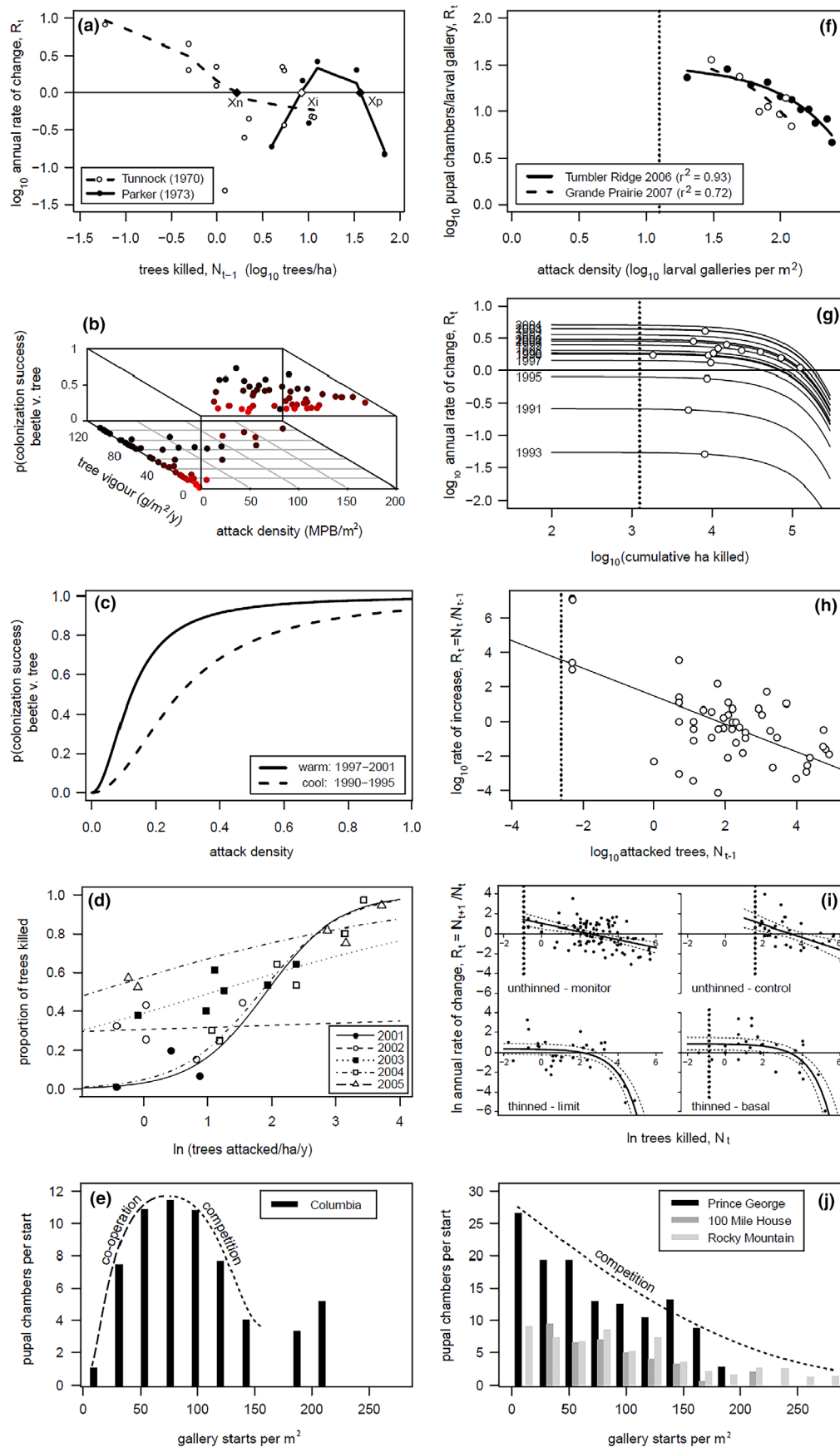


FIGURE 2 Reproduction functions for mountain pine beetle populations presented in the ten numbered studies of Table 2. The left-hand column (a)–(e) lists studies that indicate an Allee effect, signified by the rising early portion of the recruitment curve, i.e. nonlinear positive density-dependence. The right-hand column (f)–(j) lists studies that either do not report an Allee effect or note the absence of an Allee effect in the recruitment, signified by linear or nonlinear negative density-dependence. The vertical dotted line in the right-hand graphs indicate the density threshold below which an Allee effect might have been observed, but for which the necessary monitoring and sampling effort was likely absent. Table 3 explores possible reasons for this absence. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jen.13362)]

Sawtooth national recreation area

Two of the field studies were conducted in the Sawtooth National Forest of Idaho, with Heavilin and Powell (2008) analysing a 30 m x 30 m raster of tree attacks spatially, but year-by-year and Powell and Bentz (2009) analysing the cumulative area attacked using the same data, but as a longitudinal time-series of aggregated spatial data. Powell and Bentz (2009) assumed a simple negative exponential growth relationship and showed that this was sufficient to explain much of the variation in the time series, once yearly variations in temperature and their numerous effects on beetle 'effectiveness' were taken into account. Heavilin and Powell (2008) showed that although the Allee effect might have been unnecessary to explain the progression of the outbreak over time, the Allee effect did help to explain the fine-scaled spatial pattern of contagion in the study area in 8 of the 12 years of study. Although Heavilin and Powell (2008) were using aerial detection survey data, they were also using a powerful spatiotemporal statistical model parameterized with 12 years of data spanning millions of raster cells and this study was unique in that regard.

Aggregate population proxy data

An earlier investigation assembled data from nine studies that measured MPB-caused tree mortality in managed and unmanaged stands and used these data to build reproduction functions (MacQuarrie & Cooke, 2011). That study did not detect Allee effects although the epidemic portion of the function did vary with the silvicultural history of the stand. The lowest observed attack densities in the aggregate data were 0.13–0.37 trees/ha (Amman et al., 1988; fig. 3) or an order of magnitude less than Berryman's (1999) estimate for X_n and an order of magnitude greater than Amman's (1984) estimate. Regardless, the scale of those values supports our proposition that populations in the incipient epidemic phase can be surveyed to locate the one or two infested trees in a stand. Extending this argument, if observers can detect very small populations on the order of 0.13–0.37 trees/ha then they should most certainly be able to detect populations near X_i with a threshold of 2–12 trees/ha. By deduction, then, the number of trees at which reproduction peaks (b in Figure 1), must therefore be larger than 2–12 trees and thus any increase in the number of dead trees that is between X_i and b should give positive per-capita recruitment rates, R , that would be revealed in data as the rising phase of the Allee curve ($R_a < R < R_b$). Despite this, the aggregate data do not show the rising phase expected of the Allee effect (fig. 3f–j).

The expected negative sloping recruitment curve predicted for the epidemic phase was observed in almost all of the data sets examined in MacQuarrie and Cooke (2011). The only exception was

in the data from Parker (1973) and Schmid et al. (2007) where per capita recruitment rates from mortality estimates between 1 and 7 trees per ha show a positive slope. The lack of evidence in most of the data for the rising phase can be explained in three, non-mutually exclusive ways: (1) the rising phase is bypassed because of a relaxation of the Allee effect as described above (Box 1), (2) the rising phase is missed because data are collected after the population enters the epidemic phase, or (3) the rising phase can only be quantified under rare conditions that occur infrequently on the landscape. We suspect the first two reasons explain most of the data in the aggregate data set. Most of these studies began after an outbreak was known to be occurring either in the stand or nearby (see appendix to MacQuarrie & Cooke, 2011). The third explanation describes a phenomenon where the irruption occurs because a small MPB population was able to breach the unstable threshold without an accompanying saddle-node bifurcation and where the observations of that population were started before, or as soon as, the first dead tree was observed in the stand. Only the studies reported by Parker (1973) and Schmid et al. (2007) appear to meet these conditions (MacQuarrie & Cooke, 2011). This may be why part of the rising phase is visible in these data.

Attack success in detailed ground surveys

There are two ways that a transition from endemic to epidemic phases may be captured, Waring and Pitman (1985) pursued one means, while Boone et al. (2011) pursued the other. The first strategy – which substitutes space for time – is to target a large enough forest area in a single year so that both endemic and epidemic populations are being sampled. Here, an aerial survey might be used initially to detect attack clusters, but a follow-up ground survey would be used to characterise population attack rates and colonisation success. The second strategy – which substitutes time for space – is to choose a smaller forest known to harbour only endemic populations and patiently wait for an outbreak to emerge so that the endemic and epidemic population phases are captured through a longitudinal series in time. In each case, investigators must make careful recordings on each stem of the number of attacks and whether the tree was successfully colonised. In both these studies, the investigators made considerable efforts on the ground to ensure there were low-density attacks present to observe the high rate of pitch-out that often happens with low-density attacks.

Boone et al. (2011) bulked their spatial data across years. Interestingly, when the raw data were analysed year-by-year, there was a clear indication that the strength of the Allee effect varies amongst years (see Cooke & Carroll, 2017, figure A2). Howe et al. (2022) added to this dataset, showing further that this annual

variation was directly attributable to environmentally mediated shifts in MPB host selection behaviour and performance/tree vigour, much as modelled by Cooke and Carroll (2017), who suggested that this weather-dependent, co-operative establishment curve may be used as a component recruitment response to derive a full recruitment model that explains the weather-dependent irruptive dynamics of MPB in the southcentral interior of British Columbia. The model illustrates how a lowering of the incipient threshold X_i and a raising of the endemic threshold X_n may precipitate an outbreak should the condition $X_n > X_i$ ever become satisfied, for example, through a sudden drought that impairs host-tree defences or a gradual climate warming that raises overwintering larval survival.

4 | DISCUSSION

MPB populations exhibited irruptive dynamics at all spatial scales, across most of their range. Although the evidence supporting the existence of Allee-driven irruptive dynamics varies considerably amongst studies, the determining factors appear to be the type of data collected and their spatial resolution. The probability of detecting an Allee effect rises when study areas are large and are sampled early in the irruption process and intensely using ground surveys where attention is paid to the fate of low-density attacks. The ability to quantify the effect of extrinsic environmental variables on MPB irruptive potential hinges critically on this more fundamental ability to quantify the positive effect at low density of population size on host-tree colonisation success. As Howe et al. (2022) stated, 'numbers matter'. If the transition from endemic to outbreak behaviour occurs at low numbers and in exquisite response to fast-changing environmental conditions affecting host vigour, then sampling numbers and detection thresholds also matter. This provides some clear guidelines as to the data standard that must be met by any operational data stream seeking to inform modern data-driven, goal-oriented predictive pest management.

A statistical meta-analysis of the Table 2 data is not possible given the low sample size and the lack of independent replicates amongst studies; however, it appears that operational aerial surveys of tree mortality are often inadequate for detecting Allee effects, which occur at a level below the operational threshold of detectability of endemic MPB populations. Aerial survey data that are spatially aggregated into time series are typically too insensitive to low-density attacks to quantify the irruptive threshold, although they may be used to gauge peak population growth potential and collapse dynamics of post-irruptive populations. Experimental-quality population monitoring data may be adequate if the data are collected at the level of the individual tree and not aggregated across the study area and if the sampling occurs early enough in the outbreak sequence that the transition from endemic to irruptive is captured. Population monitoring efforts that employ either space-for-time substitutions (perhaps aided by aerial surveys) or time-for-space substitutions (i.e., focussed longitudinal time series) as conducted by Waring and Pitman (1985) and Boone et al. (2011), respectively, can capture the

endemic-epidemic transition because they rely on ground assessments of population attack rates and colonisation success. However, either strategy has its risks, as the environmental factors affecting population growth rates are sure to vary in both space and time. The lowest-risk strategy is, of course, to sample large areas over long time frames, but this is also costly. The optimal monitoring approach will therefore depend on the amount and duration of the funding available for the program.

Under the premise that the Allee effects should be visible in all MPB tree mortality data, our findings reveal an interesting conflict with regard to the strength of the demographic Allee effect in this species. Some recruitment curves suggest MPB is governed by a strong Allee effect (Figure 2a-e), while others might be taken to suggest there is no Allee effect (Figure 2f-j). We contend that it is the lack of quantification of the dynamics of extremely low densities of MPB that renders the recruitment curves in Figure 2f-j unusable as descriptors of the irruptive dynamics of transitions from the endemic to the epidemic state. Although they may help to describe the rate of growth of irruptive populations and the rate of collapse of post-irruptive populations, only the recruitment curves on the left capture the early phase dynamics of irruption from the endemic to the incipient stage. The curves in Figure 2f-j highlight an absence of data points to the left of the vertical bar, which we attribute to the time lag between the endemic and irruptive states that are typically involved in prompting the initiation of operational surveys. This time lag has been estimated at roughly 5 years for the average lodgepole pine forest (Safranyik & Carroll, 2006), but the time to grow from endemic to irruptive to survey could be faster or slower, depending on variations in climatic suitability. Because of this time lag in the initiation of operational surveys, we are sceptical of the validity of the extrapolations in the proxy-based recruitment curves of Figure 2f-j.

The critical endemic and irruption thresholds that are reported in the literature vary widely and the basis of this variation is unclear, but might stem from several sources, including ambiguous starting definitions, the choice of study area, the population studied, the type of data gathered, the method of data processing and the type of model fit to the data. Based on the five studies which reported an Allee effect (Figure 2a-e), the strength of the intrinsic Allee effect varies significantly in response to fluctuations in extrinsic factors, from weather-driven fluctuations in microclimatic conditions to variations in host vigour and tree defences.

The various studies (Table 3) are so idiosyncratic that they can not be considered replicates of one another. Indeed, this might be expected if each new study that is done is aiming for novelty over comparability. Future studies should attempt to clarify how their results relate to these earlier results and in doing so should attempt to explicitly address the roles of weather, climatic and microclimatic variability in shaping recruitment patterns.

An additional consideration is the importance of choosing forests that are not subject to heavy immigration of beetles from afar. If a study area is subject to heavy immigration, then the recruitment curves will not be representative of local population growth dynamics and evidence for Allee effects may be obscured or

enhanced. This may have been part of the problem with Trzcinski and Reid (2009), where an outbreak on the northeast-facing slope of the Kootenay River was likely initiated years after and in response to, the outbreak on the opposite, southwest-facing side of the river a decade earlier. This was also a likely issue with the Tumbler Ridge-Grande Prairie study, where the outbreak on the East slopes of the Rocky Mountains was fuelled by immigration from the unprecedented outbreak in northern British Columbia on the West side of the Rockies (de la Giroday et al., 2012). Multiple investigations conducted in the Sawtooth National Recreation Area of Idaho (Amman, 1984; Heavilin et al., 2007; Heavilin & Powell, 2008; Logan & Powell, 2004; Powell & Bentz, 2009, 2014) have emphasized the importance of this area as a closed arena not vulnerable to immigration. Our results support the view that heavy immigration can lead to inflated daily attack rates, thus obscuring the Allee effect that is detected only when low-density attacks are observed.

This study reveals an important ambiguity in the accepted definition of an 'endemic' MPB population. Amman (1984) defined an endemic threshold that would only be visible for plots larger than 100 acres, which is much larger than the average plot size. However, this threshold was not estimated, but rather 'guesstimated', based on the author's experience. Likewise, the definition of endemic proposed by Safranyik and Carroll (2006), although based on objective measurements, was determined by a biological consideration of beetle behaviour and not estimated through an analysis of beetle population dynamics. We have attempted to show how a dynamic endemic equilibrium state may exist well above the threshold at which beetle attack is predicted to switch from a focus on small, unhealthy host trees to large, healthy host trees. This highlights a certain ambiguity in the terms 'endemic' and 'epidemic' in the larger literature on bark beetle ecology. We propose that the definitions developed here (Box 1) can be used as a common point of reference for other studies of population dynamics in bark beetles. This would be consistent with later work by Burke and Carroll (2017) and Howe et al. (2022), who made considerable progress towards illuminating the connections between population density, individual behaviour and the population-level consequences of co-ordinated, density-dependent, environmentally mediated shifts in host-selection behaviour.

Finally, these results support the operational findings of field sampling in the province of Alberta, where the MPB has been the target of control programs every year since 2006. The outbreak in Alberta is an exceptional case where widespread, but detailed surveys were conducted annually by observers in low-altitude rotary winged aircraft, followed by ground surveys and aggressive destruction of all infestations of 3 or more trees. This produced a quantifiable slowing of the rate of spread of MPB in Alberta through the years 2006–2016 (Carroll et al., 2017). The provincial detection and control program was premised on the likelihood that low-density attacks in marginal climates would contribute little to irruptive spread, which is our primary result. The province-wide rate of detection of newly infested trees was relatively low throughout this campaign,

despite the considerable investment in detection effort, yet the program succeeded in part because the control effort was deployed judiciously (Carroll et al., 2017). Had the rates of detection been much lower, the removal rate might have been insufficient to offset the natural rate of population growth under an environmentally mediated Allee effect. The operational experience in Alberta supports our experimental observation that there is an unstable threshold below which populations fail to contribute to irruptive growth and spread and this threshold varies according to environmental conditions.

5 | CONCLUSION

Our over-arching conclusion is that MPB exhibits irruptive dynamics at all spatial scales, from the stem to the stand to the landscape as suggested by Raffa et al. (2008), right across their range. The Allee effect is ubiquitous geographically. It is strong, but it may weaken considerably under warm temperatures, which favour the beetle over the tree, thereby resulting in a classic saddle-node-bifurcation driving MPB outbreak. The endemic and incipient thresholds and maximum reproductive rates are not fixed parameters, but, rather, functions, which vary strongly in response to dynamic environmental conditions.

The reason that evidence for the irruptive Allee effect appears to wane with the largest-scale operational monitoring programs is because of the type of data that can be collected efficiently at that scale and the tendency for these aerial detection surveys to be initiated too late in the irruptive process. Tree mortality data are a reasonable proxy for population data, but only when populations are post-irruptive. For populations that are pre-irruptive, ground surveys of attack success are required to characterise the nature of the irruptive behaviour. To date, only three studies have been able to capture this effect and yet this is precisely what is required if we are to understand how low-density irruptive behaviour changes with changing environmental circumstances in space and time, including microclimatic variability, climate change and forest landscape change.

Our findings may be generalised to other bark beetle species and other systems where Allee effects are predicted to occur at different scales of observation relevant to the biology of the species. If a demographic Allee effect is predicted by the biology of the species, but not detected, then the reasons for failing to detect it should be carefully examined. It is essential that the survey methods are sensitive enough to detect very low levels of occurrence.

AUTHOR CONTRIBUTIONS

B. J. Cooke: Conceptualization; investigation; writing – original draft; methodology; visualization; writing – review and editing; software; formal analysis; project administration; data curation; supervision; resources; funding acquisition. **C. J. K. MacQuarrie:** Investigation; writing – original draft; writing – review and editing; software; formal analysis; data curation. **A. L. Carroll:** Investigation; validation; writing – review and editing; supervision.

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CONFLICT OF INTEREST STATEMENT

No, there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Research Gate Data Archive at <https://doi.org/10.13140/RG.2.2.25960.65283>.

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